MINA: Model Interrogation of Nucleotide Architectures reveals coding-sequence family signal in frozen DNA encoders

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Abstract

Motivation: Frozen DNA encoders are often used as genomic feature extractors, but fine-tuning does not show what their unchanged embeddings already contain. We introduce MINA (Model Interrogation of Nucleotide Architectures), a lightweight probing benchmark for testing whether frozen DNA embeddings encode the semantic information needed to recover two targets: a 5-way protein-family annotation, and the 1,536-d GenePT natural-language summary of each gene. We compare recoverability between coding and TSS-centred sequence contexts.

Results: In 3,244 human protein-coding genes from five families, frozen encoders recovered the family-annotation target most clearly from coding sequence. NT-v2 with concat pooling reached macro-F1 0.828 / κ 0.821, compared with 0.672 / κ 0.702 for a CDS 4-mer baseline. Alignment to GenePT natural-language descriptions was weaker. Replacing CDS with 196,608 bp TSS-centred windows collapsed performance across all four encoders, indicating that the recoverable signal is primarily coding-sequence family signal rather than generic gene-function signal from arbitrary genomic context.

Availability and implementation: Source code: https://github.com/Austin-Senna/dna_to_text; Python $\geq$3.11.

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Supplementary information: Supplementary data are available in the appendix.

1 Introduction

Pretrained DNA encoders such as DNABERT-2 (Zhou *et al.*, 2024), Nucleotide Transformer (Dalla-Torre *et al.*, 2025), GENA-LM (Fishman *et al.*, 2025), and HyenaDNA (Nguyen *et al.*, 2023) are increasingly used as genomic feature extractors. They differ in tokenisation, architecture, context length, and pretraining corpus, yet their embeddings are often treated as interchangeable summaries of biological sequence. Strong downstream performance after fine-tuning, however, does not show what information is already present in the frozen representation. Linear probing (Alain and Bengio, 2018) addresses that narrower question by asking what a simple readout can recover without changing the encoder.

The central question is whether frozen DNA representations carry gene-level functional signal beyond local nucleotide composition, and whether that signal depends on the sequence context supplied to the encoder. A canonical coding sequence contains protein-domain and family information, whereas a long window around a transcription start site samples mostly noncoding sequence around the gene's promoter. A useful representation for one substrate need not expose the same signal from the other.

This framing is distinct from recent DNA-language modelling work. Omni-DNA uses autoregressive genomic pretraining followed by multi-task fine-tuning for tasks including DNA-to-function text generation, while shared-tokenizer and other DNA-language fusion methods train alignment or integration modules over DNA-text pairs (Li *et al.*, 2025, 2026). These systems show that sequence-text modelling is feasible, but they add substantial multimodal training. GenePT instead represents each gene by embedding a natural-language summary, providing a text-derived gene vector that can be used as a cross-modal target (Chen and Zou, 2023). We use that target to test lightweight linear recoverability from frozen DNA embeddings rather than to train a new DNA-text alignment model.

Here, we introduce MINA (Model Interrogation of Nucleotide Architectures), a lightweight probing benchmark for frozen nucleotide encoders, with composition, pooling, and sequence-context controls. MINA tests four frozen self-supervised DNA encoders on two readouts: 5-way protein-family classification, and Ridge regression into the 1,536-d GenePT natural-language target. The corpus contains 3,244 human protein-coding genes from transcription-factor, GPCR, kinase, ion-channel, and immune-receptor families. CDS experiments include a 4-mer compositional baseline, shuffled-label anti-baselines, and a pooling sweep across encoder summaries. We then compare coding sequence with 196,608 bp windows centred on transcription start sites (TSS), using matched TSS 4-mers, all four encoders evaluated on the same TSS windows, and Enformer (Avsec *et al.*, 2021) as a supervised sequence-to-function comparator. The evidence we report is therefore for two specific things: the protein-family annotation recovered from CDS, and the GenePT natural-language target partially recovered from CDS; we do not claim recovery of arbitrary gene function from noncoding regulatory context.

2 Methods

Dataset. We curated 3,244 human protein-coding genes from five functional families: transcription factors (1,743), G-protein-coupled receptors (591), protein kinases (558), ion channels (198), and immune receptors (154). Genes were selected by regex match against the HGNC `gene_group` field (Seal *et al.*, 2023), intersected with the GenePT v2

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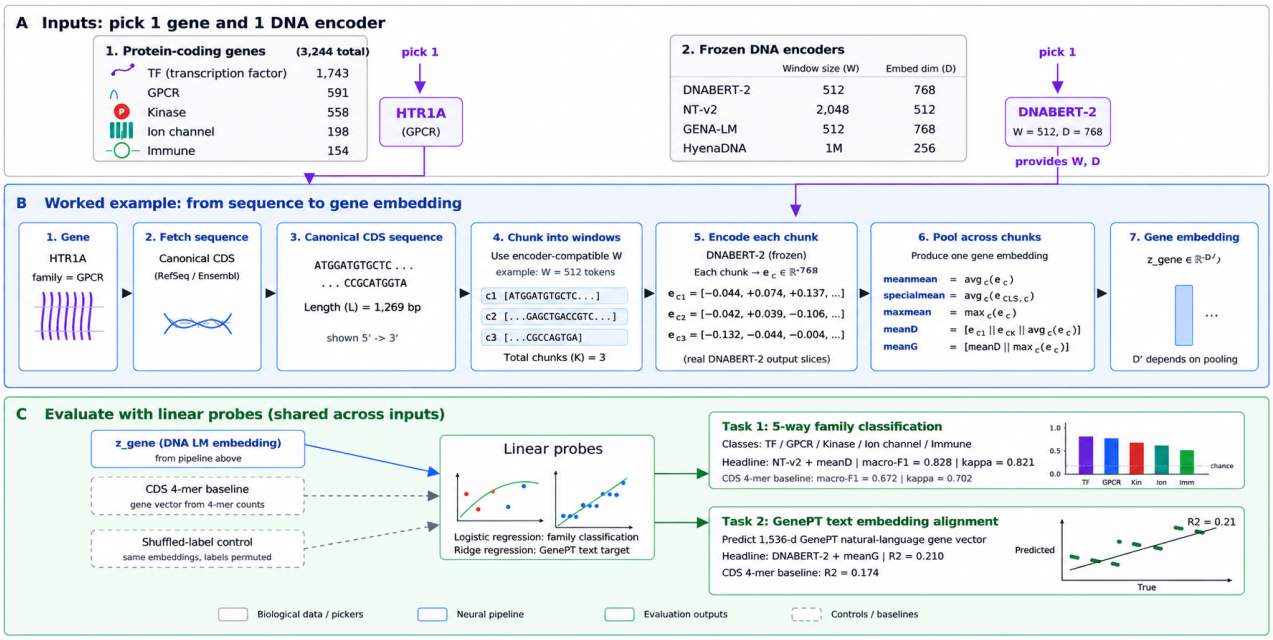

Fig. 1. Study design. Human protein-coding genes are represented by canonical CDS, encoded with frozen DNA language models, reduced to fixed gene embeddings by chunking and pooling, and evaluated with linear probes for family classification and Ridge-to-GenePT alignment. CDS 4-mer and shuffled-label controls test compositional and pipeline-leak baselines.

gene-symbol set (Chen and Zou, 2023), and deduped first-family-wins in the assignment order kinase → transcription factor → ion channel → GPCR → immune receptor (so a gene matching multiple HGNC groups is counted under the earliest-priority family; e.g., a HIPK kinase that also has a TF role is counted as a kinase). The Family column in Table 1 is therefore size-ranked but not assignment-ranked. Canonical coding sequences (CDS) were retrieved from Ensembl (Harrison *et al.*, 2024). The corpus was split 70/15/15 stratified by family with seed 42 and reused for all downstream runs (Table 1).

Of the 18,836 HGNC protein-coding genes covered by GenePT v2, 3,247 match one of the five family regexes and 3,244 of those have a retrievable Ensembl CDS, so the corpus sits at 99.9% of the within-five-families ceiling. We do not extend to the remaining ∼15,600 GenePT-covered protein-coding genes because weaker-domain families would likely shrink the encoder-versus-k-mer gap (§5.2). That scope is a separate study.

Table 1. Dataset composition and split.

Family	Total	Train	Val	Test
Transcription factor	1,743	1,220	262	261
GPCR	591	414	88	89
Kinase	558	390	84	84
Ion channel	198	138	30	30
Immune receptor	154	108	23	23

Encoders. We compared four frozen self-supervised DNA encoders on canonical CDS. **DNABERT-2** (Zhou *et al.*, 2024) is a 117 M-parameter BERT-style model (`zhihan1996/DNABERT-2-117M`) with byte-pair encoding, 768-d hidden states, a 512-token context window, and `[CLS]` / `[SEP]` boundary tokens. **Nucleotide Transformer v2 multi-species 100M** (Dalla-Torre *et al.*, 2025) (`InstaDeepAI/nucleotide-transformer-v2-100m-multi-species`, hereafter NT-v2) is an ESM-style masked-language model with fixed non-overlapping 6-mer tokens, 512-d hidden states, a 2,048-token context window, and a `<cls>` marker. **GENA-LM base** (Fishman *et al.*, 2025) (`AIRI-Institute/gena-lm-bert-base-t2t`) is a 110 M-parameter BERT-12L masked-language model with 768-d hidden states, 12 attention heads, a 32,000-token BPE vocabulary, and a 512-token context window corresponding to roughly 4.5 kb after BPE compression; the `t2t` checkpoint was trained on the human T2T assembly with 1000 Genomes SNP augmentation. **HyenaDNA large** (Nguyen *et al.*, 2023) (`LongSafari/hyenadna-large-1m-seqlen-hf`) is an attention-free causal next-nucleotide model based on Hyena sequence operators, with single-nucleotide tokens, 8 layers, 256-d hidden states, approximately 6.6 M trainable parameters, and a context window up to 1M nucleotides; HyenaDNA was pretrained on the human reference genome. All encoder weights were frozen.

Encoding and pooling. Canonical CDS were chunked to fit each encoder's context window, wrapped with model-standard boundary tokens where applicable (`[CLS]`/`[SEP]` for DNABERT-2 and GENA-LM, `<cls>` for NT-v2; HyenaDNA has no learned summary token), and encoded in PyTorch via Hugging Face Transformers (Wolf *et al.*, 2020). Because each gene must then be represented by a fixed-size vector, we evaluated simple parameter-free pooling rules rather than training an additional sequence aggregator. Mean, max, and CLS pooling are standard reductions for contextual encoder embeddings (Devlin *et al.*, 2019; Reimers and Gurevych, 2019); `specialmean` retains model boundary/special-token positions; and `meanD` and `meanG` add coarse first/last/global summaries so long genes are not represented only by an unordered average. The full cached CDS matrix for 5-way family classification and Ridge-to-GenePT regression is reported in Supplementary Table S1.

Variant	Gene-level reduction
`meanmean`	average token embeddings, then average chunks
`specialmean`	average all tokens including special tokens, then average chunks
`maxmean`	max-pool token embeddings, then average chunks
`clsmean`	use CLS-position embedding, then average chunks
`meanD`	concatenate first chunk, last chunk, and chunk mean
`meanG`	`meanD` plus a max-over-chunks summary

TSS context and Enformer comparator. Non-coding regulatory sequence can carry gene-function signal that protein-domain motifs do not: genes in the same co-regulated module share promoter and enhancer motifs (e.g., NF-κB binding sites near immune-response genes, ISRE elements near interferon-stimulated genes), and that shared regulatory structure is plausibly mirrored in natural-language descriptions of pathway co-membership. The TSS arm therefore tests two related but distinct hypotheses: whether regulatory context recovers the same discrete protein-family label as CDS, and whether it recovers the continuous GenePT natural-language target that is more directly aligned with co-regulated-function vocabulary. To test whether the same family signal is recoverable from TSS-window sequence context, we constructed 196,608 bp genomic windows centred on each gene's strand-specific transcription start site. These windows are much larger and mostly noncoding rather than protein-domain-rich, so we expected weaker family-label recovery than from CDS while gaining a direct CDS-versus-TSS comparison.

We ran all four self-supervised DNA encoders (NT-v2, DNABERT-2, GENA-LM, HyenaDNA) over the same 196,608 bp TSS windows with the CDS pooling rules and probe protocol, plus a matched TSS-window 4-mer baseline; for each encoder we report the best-pool result against the TSS 4-mer baseline. We also included Enformer (Avsec *et al.*, 2021) as a supervised TSS-to-function comparator rather than a CDS encoder. Enformer takes 196,608 bp DNA input and predicts functional genomic tracks over 896 central 128-bp bins, with 5,313 human and 1,643 mouse output tracks, making it a natural baseline for long TSS-window sequence features. From each Enformer run we computed three summaries for the same probe recipe: `trunk_global` (mean across all 896 bins), `trunk_center` (mean across the central 16 bins around the TSS, 2,048 bp total), and `tracks_center` (mean predicted track values over the same central bins).

Probes. For classification we trained multinomial logistic regression. For the 1,536-d GenePT text-embedding target we trained multi-output ridge regression with `scikit-learn` (Pedregosa *et al.*, 2018), on top of `NumPy` (Harris *et al.*, 2020) and `SciPy` (Virtanen *et al.*, 2019); ridge was chosen over reduced-rank or partial-least-squares variants for its closed-form efficiency on the high-dimensional target and its alignment with the standard linear-probing protocol. Regularisation (α for ridge, C for logistic) was swept on the validation split over a decade grid, the probe was refit on the train+val concatenation at the selected hyperparameter, and metrics were evaluated *once* on the held-out test split; selected hyperparameters are reported per cell in Supplementary Table S1. Linear probing follows the protocol of Alain and Bengio (2018).

Baselines and controls. The *k-mer composition* baseline projects each CDS, or each TSS window for the context ablation, to a 256-d L_1-normalised 4-mer histogram and runs the same probe recipe; it is encoder-independent and serves as the compositional floor. The *shuffled-label anti-baseline* permutes family labels in train+val only (test labels remain real) and tests pipeline honesty: chance-corrected agreement should collapse to zero. For the regression probes, an analogous *shuffled-Y anti-baseline* permutes the 1,536-d GenePT vectors in train+val only and re-runs the full α sweep on the shuffled validation split (rather than reusing the α selected from the real run), so any non-trivial test R^2 would imply leakage in either the data split or the hyperparameter selection.

Evaluation metrics. For 5-way family classification we report macro-F1 and Cohen's κ (Cohen, 1960) jointly: macro-F1 weights each family equally and is appropriate for the imbalanced family distribution, while κ is the chance-corrected agreement metric. The two carry different information: κ rewards majority-class agreement under imbalance, whereas macro-F1 rewards minority-class recovery, so we report both for every classification cell. Accuracy and confusion matrices are reported as supporting diagnostics; classification deltas are reported relative to the CDS or TSS 4-mer baseline as appropriate. For Ridge-to-GenePT regression we report macro-averaged held-out R^2 across the 1,536 GenePT dimensions and mean cosine similarity between predicted and target GenePT vectors, with R^2 deltas relative to the CDS 4-mer baseline. For headline cells we report 95% confidence intervals from a 1,000-iteration bootstrap of the held-out test split (stratified by family for classification, i.i.d. by gene for regression); these intervals quantify test-composition sampling uncertainty, not hyperparameter or split-seed variability.

3 Results

3.1 CDS encoders separate gene families

On canonical CDS, NT-v2 is the strongest encoder for 5-way family classification: `meanD` pooling reaches macro-F1 0.828 (95% bootstrap CI [0.77, 0.87]) / Cohen's κ 0.821 ([0.77, 0.86]), a $\Delta\kappa$ of +0.119 over the CDS 4-mer baseline (macro-F1 0.672 [0.60, 0.72] / κ 0.702 [0.64, 0.75]; Table 2). The gain is not just majority-family recovery: per-class F1 improves most for the minority immune and ion-channel classes ($+0.33$ and $+0.29$ over 4-mers). The shuffled-label anti-baseline collapses to macro-F1 0.208 / κ 0.048, so the family signal is not split leakage or probe bookkeeping. HyenaDNA is competitive with DNABERT-2, while GENA-LM base falls below CDS 4-mer composition; thus, "DNA language model" is not a sufficient explanation for success.

Table 2. Best cached 5-way family classification cell. Best non-control value is bolded.

Source	Pool	F1	κ	$\Delta\kappa$	Acc.
Shuffled labels	—	0.208	0.048	-0.654	0.435
CDS 4-mer	—	0.672	0.702	0.000	0.817
DNABERT-2	`meanD`	0.738	0.723	+0.020	0.823
NT-v2	`meanD`	**0.828**	**0.821**	**+0.119**	**0.887**
GENA-LM base	`clsmean`	0.498	0.478	-0.225	0.686
HyenaDNA large	`meanG`	0.715	0.694	-0.008	0.809

Test split metrics; full pooling sweep in the appendix.

Figure 2 shows where the family5 errors concentrate for the headline NT-v2 `meanD` cell. Figure 5 (§3.5) visualises the same embedding under UMAP, where the five families form visually-separable regions.

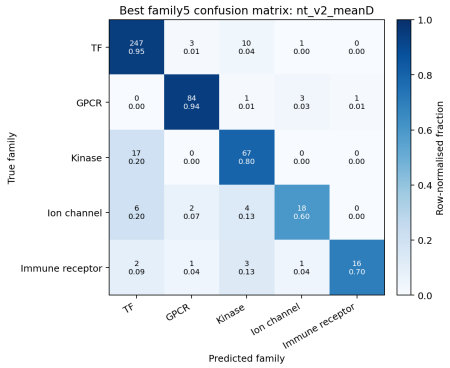

Fig. 2. Row-normalised confusion matrix for the headline 5-way family classification cell, NT-v2 `meanD`. Diagonal mass concentrates in TF and GPCR; the largest off-diagonal errors are kinase-as-TF and ion-as-TF, consistent with the minority-class noise in Table 2.

3.2 Cross-modal alignment is weaker than family classification

Linear projection into the 1,536-d GenePT text-embedding space recovers a smaller signal than family classification (Table 3). DNABERT-2 `meanG` is the best Ridge probe (R^2 0.210, 95% bootstrap CI [0.19, 0.23]), only +0.036 above the CDS 4-mer baseline (R^2 0.174 [0.15, 0.19]). NT-v2 reaches R^2 0.193, HyenaDNA 0.182, and GENA-LM 0.117, so the strongest CDS family separator is not necessarily the representation most aligned with text-summary geometry.

Cosine similarity is much less discriminative: the shuffled-Y anti-baseline still reaches 0.913, and all real feature sets fall in a narrow 0.925–0.934 band. This compression is consistent with anisotropy in the GenePT embedding space, where many predicted vectors share a common direction. Thus, cosine is useful as a sanity check, but R^2 carries the meaningful separation among GenePT probes.

Table 3. Best cached Ridge-to-GenePT cell. Best non-control value is bolded.

Source	Pool	R^2	Δ	Cos.
Shuffled labels	—	-0.000	-0.175	0.913
CDS 4-mer	—	0.174	0.000	0.931
DNABERT-2	`meanG`	**0.210**	+0.036	**0.934**
NT-v2	`meanmean`	0.193	+0.019	0.932
GENA-LM base	`meanmean`	0.117	-0.057	0.925
HyenaDNA large	`specialmean`	0.182	+0.008	0.931

Δ is relative to the CDS 4-mer baseline; base/`meanmean` ties are shown as `meanmean`.

The modest R^2 values constrain the biological claim: GenePT summaries are heterogeneous and not limited to protein-family membership. They are also broadly rather than narrowly carried: for DNABERT-2 `meanG` CDS, 1,527 of the 1,536 GenePT dimensions have positive test R^2, the median per-dim R^2 is 0.190 (within 0.02 of the 0.210 macro mean), and recovering 50% of the summed R^2 requires the top 429 dimensions (28% of the embedding space); the other three encoders show the same broad shape one band lower (Supplementary Figure A1). The headline cross-modal signal is therefore distributed across most of the GenePT space rather than concentrated in a handful of well-predicted summary axes. The joint view in Figure 3 makes the encoder tradeoff explicit: NT-v2 is the strongest family classifier, whereas DNABERT-2 is most Ridge-aligned.

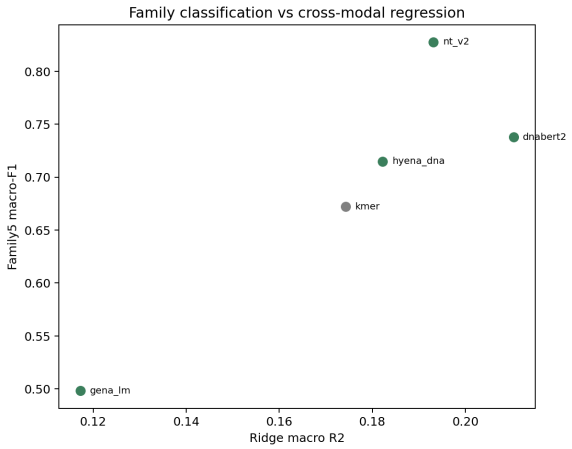

Fig. 3. 5-way family classification versus Ridge-to-GenePT alignment. NT-v2 is strongest for classification; DNABERT-2 is strongest for Ridge.

3.3 Pooling controls representation geometry

Pooling choice changes apparent representation quality (Figure 4). The heatmap uses macro-F1 because the sweep compares 5-way classification under class imbalance, and macro-F1 weights each family equally. NT-v2 spans macro-F1 0.586 (`maxmean`) to 0.828 (`meanD`); HyenaDNA's `clsmean` collapses to macro-F1 0.140 / κ 0 because the causal model has no BERT-style trained summary token, while its concat-style `meanG` reaches 0.715. Concat-style summaries (`meanD` and `meanG`) help both NT-v2 and DNABERT-2, whereas `specialmean` is nearly indistinguishable from `meanmean` on CDS. The full encoder × pooling matrix is in Supplementary Table S1; Tables 2 and 3 report best-cell-per-encoder selections from this 24-cell sweep. The model-specific pattern is consistent with each encoder's training interface: masked-language models expose boundary or summary positions, whereas HyenaDNA is causal and has no trained CLS token. Pooling is therefore part of the representation interface, not a neutral implementation detail.

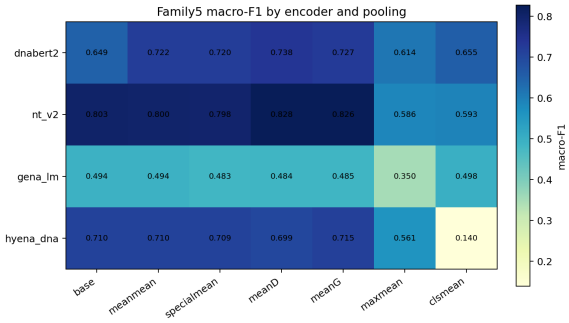

Fig. 4. 5-way family classification macro-F1 across cached encoder-pooling cells. Concat-style summaries help NT-v2 and DNABERT-2; special-token-inclusive mean pooling is largely neutral; CLS-style pooling is architecture-dependent.

3.4 Genomic context controls recoverable family signal

Holding the gene set fixed and replacing the CDS substrate with 196,608 bp TSS-centred windows collapses both readouts across every self-supervised encoder we tested (Table 4). Each encoder's classification κ drops by roughly half between CDS and TSS, and each encoder's R^2 drops to roughly half its CDS value; the substrate effect is encoder-general rather than an NT-v2 artefact. Every encoder still beats the matched TSS 4-mer baseline on both metrics, so the TSS windows are not uninformative; they expose family-correlated promoter or cis-regulatory motifs rather than independent recovery of protein-family information. Supervised Enformer trunk features improve further on TSS (κ 0.439; R^2 0.142) consistent with its supervised long-window training objective, but remain below the CDS encoders on family labels. The benchmark therefore supports a narrower claim: frozen DNA encoders recover coding-sequence family information, with recoverability controlled by genomic substrate uniformly across architectures.

Table 4. Substrate ablation: classification (κ) and Ridge-to-GenePT (R^2) per encoder, grouped by sequence context. Per-context deltas are relative to the within-context 4-mer composition baseline. NT-v2 rows are bolded as the headline encoder.

Source	κ	$\Delta\kappa$	R^2	ΔR^2
Coding sequence (CDS)				
4-mer	0.695	0.000	0.174	0.000
NT-v2	**0.815**	**+0.120**	**0.193**	**+0.019**
DNABERT-2	0.723	+0.027	0.210	+0.036
HyenaDNA	0.702	+0.007	0.182	+0.008
GENA-LM	0.480	-0.215	0.117	-0.057
TSS-centred window (196,608 bp)				
4-mer	0.207	0.000	0.041	0.000
NT-v2	**0.375**	**+0.169**	**0.117**	**+0.076**
DNABERT-2	0.320	+0.113	0.122	+0.080
HyenaDNA	0.328	+0.121	0.085	+0.044
GENA-LM	0.275	+0.068	0.059	+0.017
Enformer trunk	0.439	+0.232	0.142	+0.101

Encoder rows use each encoder's best classification pool for κ (Tables 2, A1) and best regression pool for R^2 (Table 3). Deltas are relative to the within-context 4-mer baseline.

3.5 Family clusters in the embedding space

Figure 5 shows the NT-v2 `meanD` embeddings reduced to two dimensions by UMAP (McInnes *et al.*, 2020), coloured by protein family. The five families form visually-separable regions: TFs and GPCRs occupy the cleanest clusters, while kinase, ion-channel, and immune-receptor genes show more cross-family overlap, consistent with the per-class minority noise visible in the confusion matrix (Figure 2).

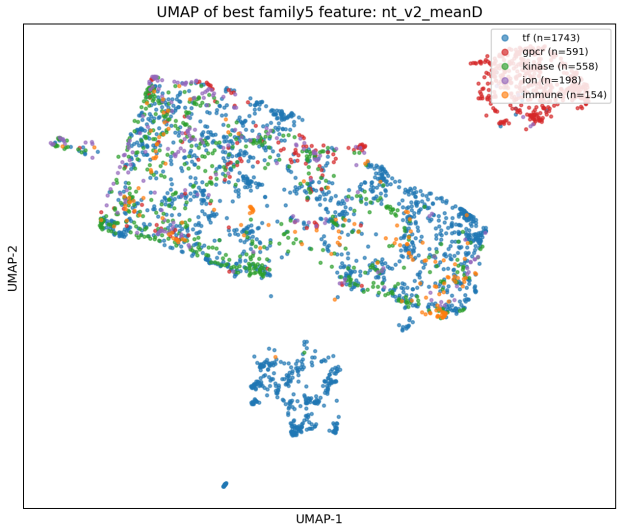

Fig. 5. UMAP (McInnes *et al.*, 2020) projection of the NT-v2 `meanD` embedding, coloured by protein family. TFs and GPCRs form the tightest clusters; kinase, ion-channel, and immune-receptor regions overlap, mirroring the off-diagonal pattern in Figure 2.

4 Discussion

Supported claim. MINA shows that frozen DNA encoders can expose coding-sequence family signal to a lightweight linear probe. The result is not a trivial success against chance: the CDS 4-mer baseline is already strong, and the shuffled-label anti-baseline collapses as expected. That strong 4-mer floor is itself biologically meaningful, because coding sequences already carry family information through nucleotide composition, codon and amino-acid biases, and short motif structure. NT-v2 nevertheless improves over CDS 4-mers by $\Delta\kappa$ +0.119 on 5-way family classification, indicating family-discriminative structure beyond unordered short-motif composition. The supported claim is therefore narrow: linearly recoverable family signal in canonical CDS, not general gene-function prediction.

Genomic context. The CDS-versus-TSS ablation is the main guardrail on interpretation because it holds the gene set fixed while changing the input substrate. Across all four self-supervised encoders run on the same 196,608 bp TSS windows the pattern is consistent: every encoder drops by 0.27–0.38 macro-F1 versus its CDS result, and all four cluster in macro-F1 0.39–0.46 on TSS with mutually-overlapping 95% bootstrap CIs. That collapse is biologically coherent: the labels are mostly protein-family labels, whereas TSS windows contain promoter, untranslated, intronic, and broader genomic sequence around each gene. Every encoder still beats the TSS 4-mer baseline (0.247) with non-overlapping CIs, so these windows are not uninformative; they carry indirect family correlates but do not match CDS encoders on protein-family labels. The substrate, not the encoder, is the dominant variable across all four self-supervised DNA encoders, ruling out an NT-v2-specific artefact. Enformer trunk features remain the strongest TSS comparator at macro-F1 0.545, consistent with its supervised long-window training objective.

Model and pooling specificity. The encoder comparison argues against treating DNA language models as interchangeable feature extractors. NT-v2 is the clearest classification winner, DNABERT-2 is strongest for Ridge-to-GenePT regression, HyenaDNA is competitive on family classification, and GENA-LM base falls below CDS 4-mer composition on both probes. Thus, pretraining alone does not guarantee a linearly useful gene representation on this benchmark. Pooling is also part of the representation interface: concat-style summaries help NT-v2 and DNABERT-2, CLS-style pooling collapses for HyenaDNA, and `maxmean` hurts every encoder tested. A plausible interpretation is that pooling works best when it matches what training made available: BERT-style models expose boundary or summary positions, concat summaries preserve coarse first/last/global information across long CDS chunks, and HyenaDNA has no trained CLS token. These patterns show why frozen-encoder comparisons need explicit pooling controls rather than a single default summary vector.

Cross-modal signal. Ridge regression into GenePT space gives a complementary but weaker view of the same representations. The best cross-modal gain is modest (ΔR^2 about +0.04 over CDS 4-mers), and the model ranking differs from classification: DNABERT-2 aligns best with GenePT summary geometry while NT-v2 best separates families. GenePT vectors are an inherently noisy target for a linear sequence probe because they inherit the geometry of English gene summaries, including shared linguistic style, broad biomedical semantics, and many features not directly determined by coding sequence. That shared structure can make vectors point in similar directions, producing high cosine even for shuffled targets, while the sequence-relevant axes remain harder to recover linearly. The compressed cosine values therefore suggest that GenePT-space alignment should be interpreted through R^2 rather than raw directional similarity. They also suggest that stronger DNA-to-text alignment may require subspace alignment or a transformed GenePT target that behaves more like a gene-vector space: preserving function-relevant axes while discarding language-embedding variation that is noisy for sequence probing.

5 Conclusion and Study Limitations

5.1 Conclusion

Frozen DNA encoders can recover coding-sequence family signal beyond simple composition, but the strength and meaning of that signal depend on genomic context, model recipe, pooling, and the target used to define "function." MINA therefore supports a precise conclusion: frozen DNA representations can be useful linearly readable CDS feature extractors, but these results should not be read as generic recovery of gene function from arbitrary genomic sequence.

5.2 Study Limitations and Next Steps

Several limitations follow directly from the design. The corpus is human-only, spans $\sim$3,200 genes, and uses five broad protein-family categories with strong domain structure; families with weaker sequence signatures may yield smaller encoder-versus-k-mer gaps. The current 3,244 genes already cover 99.9% of the within-five-families GenePT$\cap$HGNC ceiling (Methods, §Dataset), so the natural next step would be to extend the corpus to all $\sim$18,836 GenePT-covered protein-coding genes; we treat that scope as a separate study because it replaces the five-family framing with a new ontology and tests a distinct question. Family membership was deduplicated first-family-wins across overlapping HGNC groups, and the split was random-stratified at the gene level rather than clustered by sequence similarity, so paralogous genes can appear on both sides of the split. The TSS context arm covers all four self-supervised encoders plus the supervised Enformer comparator, but is restricted to a single 196,608 bp window centred on the strand-specific TSS, and Enformer's pretraining tracks include many of the gene loci it is evaluated on, so the substrate-dominance claim is constrained to this window choice and to Enformer-overlapping loci. HyenaDNA is also substantially smaller than the BERT-style encoders (256-d hidden, $\sim$6.6 M parameters), so its competitive but lower headline result mixes architecture, capacity, and pretraining objective.

On the analysis side, the headline 1,000-iteration bootstrap confidence intervals quantify held-out test-composition sampling only; they do not cover split-seed variance, hyperparameter uncertainty, or multiplicity from best-cell-per-encoder reporting across the 24-cell pooling grid. Minority-class test sets are also small (immune n=23, ion n=30), making per-class components noisy. Stronger follow-up studies would use paralog-aware splits (e.g., MMseqs2 or CD-HIT clustering at a stated identity threshold), additional organisms and label sets, broader noncoding targets, per-layer probes, paired permutation tests, and matched context sweeps across architectures and TSS window sizes.

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A Supplementary Tables and Figures

The appendix reports the full pooling sweep, the full Ridge cell grid, and the per-dimension R^2 diagnostic that support the main-text best-cell-per-encoder summary. All tables use the frozen 70/15/15 split and the same validation-tuned probe protocol as the main text.

A.1 Pooling and Regression Full Matrices

Table A1. All cached encoder-pooling cells for 5-way family classification.

Encoder	Pooling	Macro-F1	κ	$\Delta\kappa$	Accuracy
DNABERT-2	base	0.6490	0.6356	-0.0668	0.7721
DNABERT-2	`meanmean`	0.7220	0.7091	+0.0067	0.8172
DNABERT-2	`specialmean`	0.7205	0.7051	+0.0027	0.8152
DNABERT-2	`meanD`	0.7380	0.7226	+0.0202	0.8234
DNABERT-2	`meanG`	0.7275	0.7066	+0.0042	0.8131
DNABERT-2	`maxmean`	0.6144	0.5817	-0.1207	0.7331
DNABERT-2	`clsmean`	0.6547	0.6616	-0.0408	0.7906
NT-v2	base	0.8031	0.7984	+0.0960	0.8727
NT-v2	`meanmean`	0.7997	0.7982	+0.0958	0.8727
NT-v2	`specialmean`	0.7977	0.7985	+0.0961	0.8727
NT-v2	`meanD`	0.8275	0.8214	+0.1190	0.8871
NT-v2	`meanG`	0.8257	0.8179	+0.1155	0.8850
NT-v2	`maxmean`	0.5865	0.6099	-0.0925	0.7639
NT-v2	`clsmean`	0.5927	0.5689	-0.1335	0.7269
GENA-LM	base	0.4940	0.4568	-0.2456	0.6550
GENA-LM	`meanmean`	0.4940	0.4568	-0.2456	0.6550
GENA-LM	`specialmean`	0.4829	0.4589	-0.2435	0.6674
GENA-LM	`meanD`	0.4843	0.4668	-0.2356	0.6653
GENA-LM	`meanG`	0.4853	0.4899	-0.2125	0.6961
GENA-LM	`maxmean`	0.3502	0.2325	-0.4699	0.5175
GENA-LM	`clsmean`	0.4982	0.4776	-0.2248	0.6858
HyenaDNA	base	0.7103	0.6883	-0.0141	0.8049
HyenaDNA	`meanmean`	0.7103	0.6883	-0.0141	0.8049
HyenaDNA	`specialmean`	0.7095	0.6925	-0.0099	0.8070
HyenaDNA	`meanD`	0.6988	0.6830	-0.0194	0.8029
HyenaDNA	`meanG`	0.7149	0.6944	-0.0080	0.8090
HyenaDNA	`maxmean`	0.5614	0.5315	-0.1709	0.7105
HyenaDNA	`clsmean`	0.1396	0.0000	-0.7024	0.5359

Source: `analysis/tables/pooling_sweep_family5.csv`.

Table A2. All cached Ridge cells with delta versus the CDS 4-mer baseline.

Feature source	Pooling	R^2 macro	Δ vs 4-mer	Mean cosine	α
Shuffled labels	–	-0.0004	-0.1746	0.9130	1000
CDS 4-mer	baseline	0.1743	0.0000	0.9306	0.01
DNABERT-2	base	0.1812	+0.0070	0.9313	10
DNABERT-2	`meanmean`	0.2029	+0.0286	0.9333	10
DNABERT-2	`specialmean`	0.2029	+0.0287	0.9333	10
DNABERT-2	`meanD`	0.2100	+0.0357	0.9340	10
DNABERT-2	`meanG`	0.2104	+0.0361	0.9340	10
DNABERT-2	`maxmean`	0.1606	-0.0136	0.9293	100
DNABERT-2	`clsmean`	0.1911	+0.0168	0.9322	100
NT-v2	base	0.1926	+0.0183	0.9324	10
NT-v2	`meanmean`	0.1932	+0.0189	0.9324	10
NT-v2	`specialmean`	0.1931	+0.0188	0.9324	10
NT-v2	`meanD`	0.1882	+0.0139	0.9319	100
NT-v2	`meanG`	0.1902	+0.0159	0.9321	100
NT-v2	`maxmean`	0.1355	-0.0387	0.9271	100
NT-v2	`clsmean`	0.1172	-0.0571	0.9251	100
GENA-LM	base	0.1173	-0.0569	0.9251	100
GENA-LM	`meanmean`	0.1173	-0.0569	0.9251	100
GENA-LM	`specialmean`	0.1124	-0.0619	0.9246	100
GENA-LM	`meanD`	0.1084	-0.0658	0.9244	100
GENA-LM	`meanG`	0.1118	-0.0624	0.9245	1000
GENA-LM	`maxmean`	0.0400	-0.1342	0.9178	1000
GENA-LM	`clsmean`	0.1093	-0.0650	0.9242	100
HyenaDNA	base	0.1822	+0.0079	0.9313	1
HyenaDNA	`meanmean`	0.1822	+0.0079	0.9313	1
HyenaDNA	`specialmean`	0.1822	+0.0080	0.9313	1
HyenaDNA	`meanD`	0.1818	+0.0075	0.9314	1
HyenaDNA	`meanG`	0.1792	+0.0049	0.9310	10
HyenaDNA	`maxmean`	0.1329	-0.0414	0.9266	100
HyenaDNA	`clsmean`	-0.0015	-0.1758	0.9129	0.01
TSS 4-mer	baseline	0.0413	-0.1329	0.9173	0.01
TSS NT-v2	`meanmean`	0.1174	-0.0569	0.9252	0.1
TSS NT-v2	`meanD`	0.0545	-0.1197	0.9190	10
TSS NT-v2	`meanG`	0.0605	-0.1138	0.9197	10
Enformer trunk	global	0.1389	-0.0353	0.9271	100
Enformer trunk	centre	0.1425	-0.0318	0.9275	1000
Enformer tracks	centre	0.0135	-0.1607	0.9171	1000

Source: `analysis/tables/regression_full.csv`. TSS and Enformer deltas are shown relative to the CDS 4-mer baseline to match the generated artifact.

A.2 Supporting Visual Diagnostics

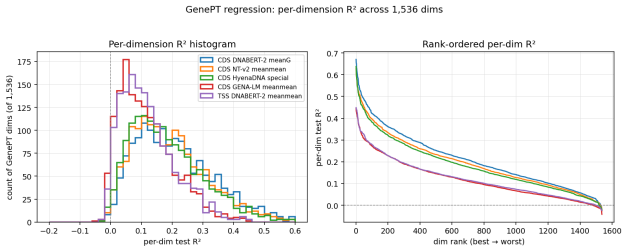

Fig. A1. Per-dimension test R^2 across the 1,536 GenePT output dimensions, for the four best CDS Ridge cells plus DNABERT-2 on TSS. Left: histogram of per-dim R^2. Right: rank-ordered per-dim R^2 (best dim on the left, worst on the right). For DNABERT-2 `meanG` CDS (macro-R^2 0.210), 1,527/1,536 dims have positive test R^2, the median per-dim R^2 is 0.190, and recovering 50% of the summed R^2 requires the top 429 dimensions (28% of the embedding space); NT-v2 `meanmean` CDS (macro-R^2 0.193) is essentially the same shape one band lower. The headline cross-modal R^2 is therefore distributed rather than concentrated in a handful of well-predicted summary axes. Source: `scripts/per_dim_r2.py` → `data/per_dim_r2.json`.

B Code Reproducibility

The project repository is available at `github.com/Austin-Senna/dna_to_text`. The repository `README.md` is the canonical entry point for reproducing the analysis. It separates the workflow into six stages: (1) gene curation and canonical CDS retrieval; (2) frozen CDS encoder extraction and pooling-dataset construction; (3) frozen split creation plus family-classification and Ridge-to-GenePT probes; (4) TSS-window construction, NT-v2 TSS extraction, and Enformer comparator features; (5) bootstrap confidence intervals; and (6) report table and figure generation. Large intermediate caches, including raw sequence caches, per-gene encoder reductions, Enformer windows, and source GenePT artifacts, are generated locally and ignored by git; report-facing metrics, tables, figures, split metadata, and selected feature/probe caches are tracked.

B.1 Reviewer-facing reproduction path

The fastest report-supporting reproduction path starts from the tracked caches and does not rerun the expensive encoder extraction. From the repository root, the README gives:

- Stage 5: run `scripts/bootstrap_test_uncertainty.py` to regenerate the 1,000-iteration bootstrap confidence intervals in `data/bootstrap_metrics.json`;
- Stage 6: run `scripts/build_analysis_artifacts.py` with `--overwrite` to regenerate report-facing tables, figures, and the artifact manifest from tracked metrics and cached outputs;
- Smoke check: run the same artifact builder with `--skip-umap`, `--overwrite`, and an output directory such as `/tmp/dna_analysis_smoke`.

The full public-source rebuild uses the same README stage order: Stage 1 builds the curated gene table and fetches CDS; Stage 2 runs the four frozen DNA encoders and materializes pooling datasets; Stage 3 trains probes and baselines; Stage 4 derives TSS-centred windows and runs NT-v2/Enformer comparisons; Stages 5–6 regenerate uncertainty estimates and report artifacts. The optional Enformer branch requires installing the repository's `.[enformer]` extra.

B.2 Deterministic settings and hyperparameters

The split and probe settings are fixed in code rather than chosen ad hoc during report writing. `scripts/make_splits.py` calls `src/splits/make_splits.py` with seed 42, stratifying on the `family` column. The resulting `data/splits.json` contains 2,270 train genes, 487 validation genes, and 487 held-out test genes from `data/dataset.parquet`. All downstream probe scripts read this same split file.

Classification probes use multinomial logistic regression with the scikit-learn `lbfgs` solver, `max_iter=2000`, and an L2 regularisation sweep over $\{10^{-2}, 10^{-1}, 1, 10, 100, 1000\}$ for C. The selected C maximises validation macro-F1. Ridge-to-GenePT probes sweep $\alpha \in \{10^{-2}, 10^{-1}, 1, 10, 100, 1000\}$, selecting by validation mean cosine. After selection, both probe families are refit on train+validation and evaluated once on the held-out test split. Shuffled-label and shuffled-GenePT anti-baselines permute train+validation targets only, using seed 42, while test labels or targets remain real.

Bootstrap confidence intervals are regenerated by `scripts/bootstrap_test_uncertainty.py` with default `--n-iters` 1000 and `--seed` 42. Classification resampling is stratified by family; regression resampling is i.i.d. by gene. These intervals therefore quantify held-out test-composition sampling uncertainty only, not split-seed, hyperparameter, or encoder-extraction variability.

Encoder extraction is deterministic up to backend numerical differences and is run with frozen model weights in evaluation/inference mode. The default `--device auto` selects CUDA, then Apple Silicon MPS, then CPU. The registered CDS chunk settings are: DNABERT-2, 510 content tokens with stride 64; NT-v2, 998 content tokens with stride 64; GENA-LM, 510 content tokens with stride 64; and HyenaDNA, 8,192 content tokens with stride 512. Pooling datasets are generated from cached per-chunk reductions for `meanmean`, `specialmean`, `maxmean`, `clsmean`, `meanD`, and `meanG` where supported by the encoder. TSS windows are 196,608 bp, centred on the strand-aware Ensembl transcription start site. For Enformer summaries, the default central-window setting averages the central 16 128-bp bins, i.e. 2,048 bp around the TSS. UMAP diagnostics use `random_state=42`, `n_neighbors=15`, and `min_dist=0.1`.